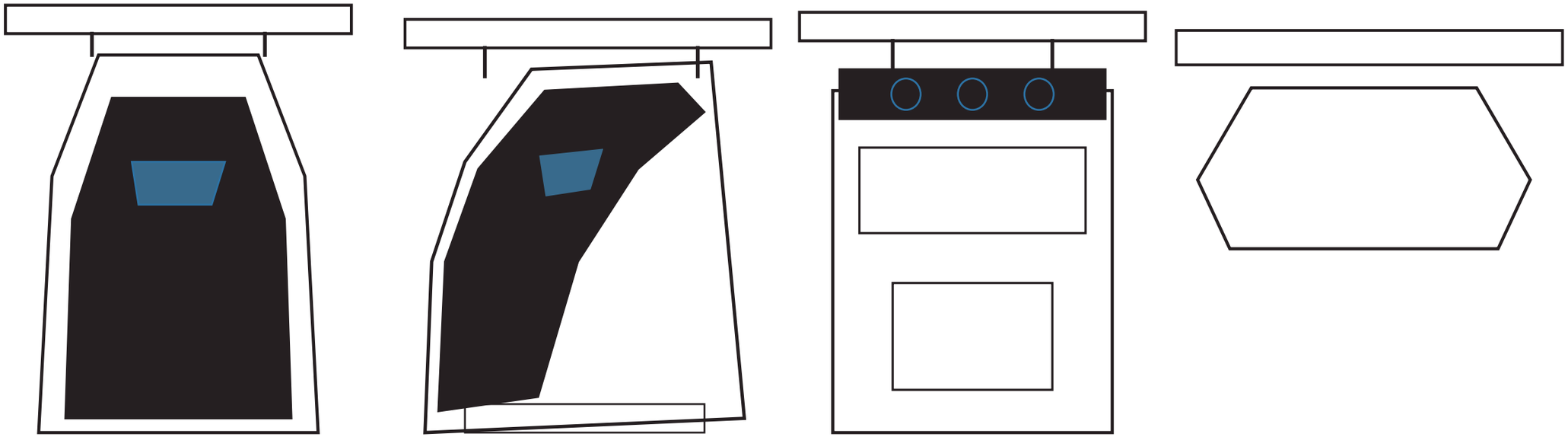
Physical mockup orthographic views

Seven supplied photos. Geometry in black is observed; blue dimensions are photo-derived estimates.



FRONT SIDE REAR TOP

Source geometry

Shell, canopy, window, top bridge, rear bays, vents and service doors.

No measurement scale

Proportions inferred across seven uncalibrated images.

None on A01

This sheet avoids tube, guideway and clinical overlays.

Estimated replica dimensions

Visual reconstruction envelope and primary features. All dimensions require confirmation on the physical mockup.

DIMENSION BASIS

No ruler, tape, photogrammetry target, known component size or camera calibration was supplied.

Main body

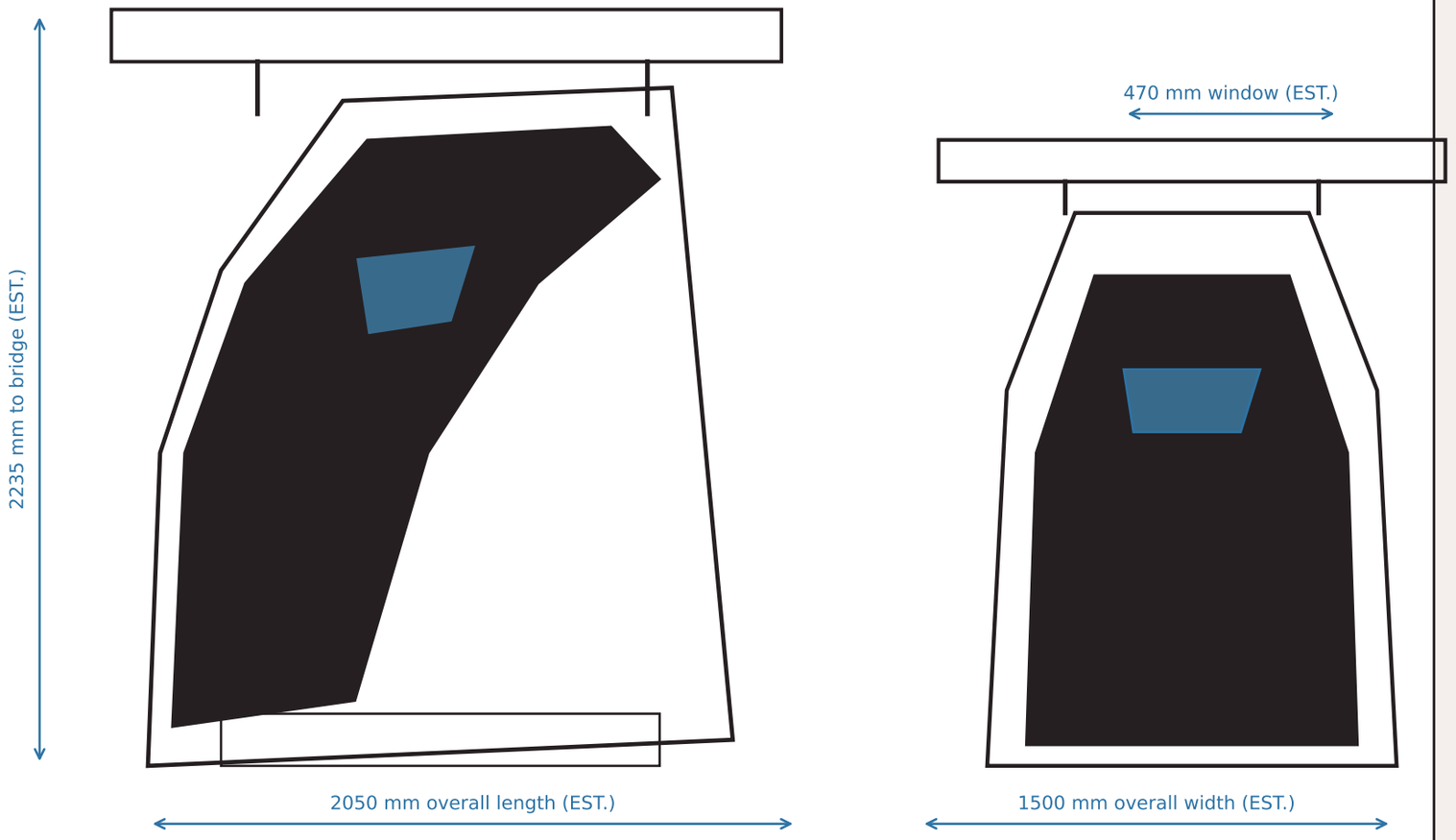
1830 L x 1200 W x 1900 H mm.

Top bridge

1500 W x 430 D x 65 T mm.

Fan / pod sizes

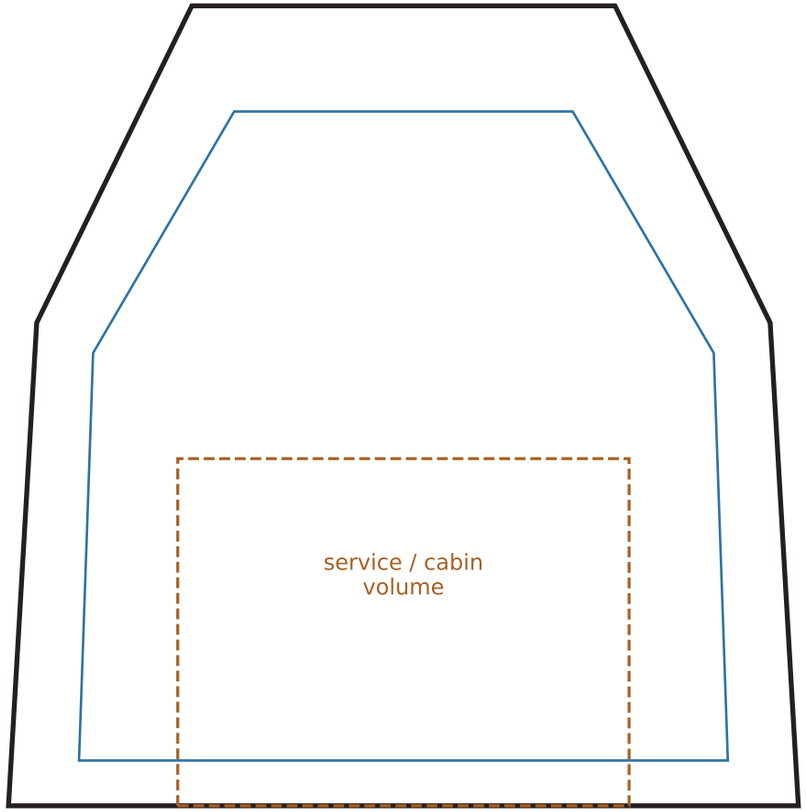
Rear fan 156 mm dia; side pod 150 mm dia.



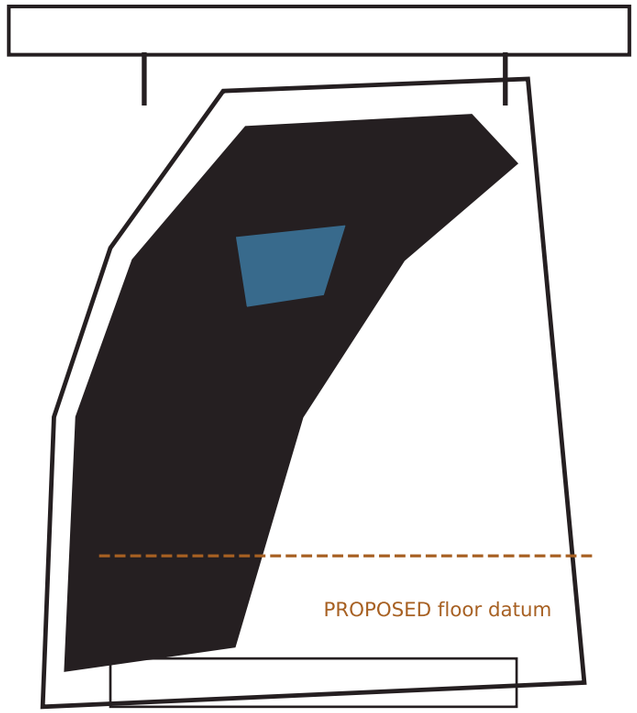
Key sections and construction reading

Observed shell interpreted as plywood skins/panels around an open service volume. Thicknesses are estimated.

SECTION A-A | TRANSVERSE



SECTION B-B | LONGITUDINAL



Outer shell

Plywood-colored side skins and black canopy.

Panel thickness

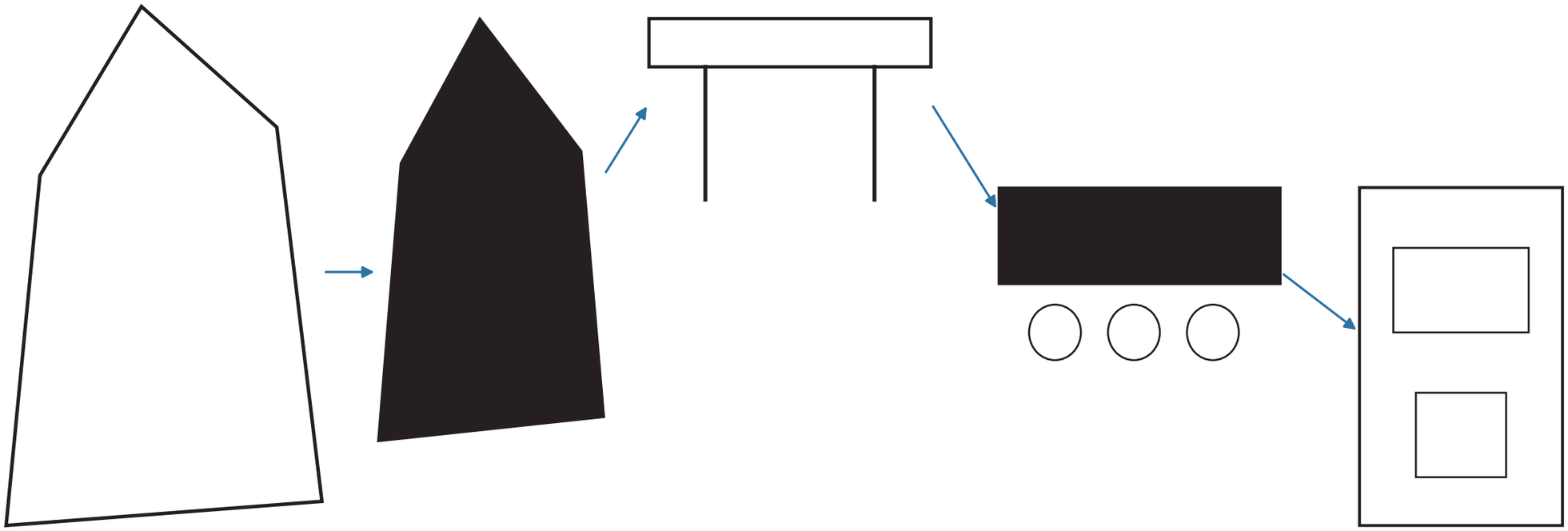
Plywood 30 mm; canopy skin 18 mm.

Cabin datum

Flat interior datum for later ambulance integration.

Exploded physical assembly

Subsystem decomposition matches the editable STL package. Arrow direction is illustrative, not an assembly sequence.



1 Shell +
white panels

2 Canopy +
window

3 Top bridge
+ supports

4 Rear bays
+ fans

5 Service doors
+ handles

Decomposition

Features grouped from visible construction seams and housings.

Interfaces

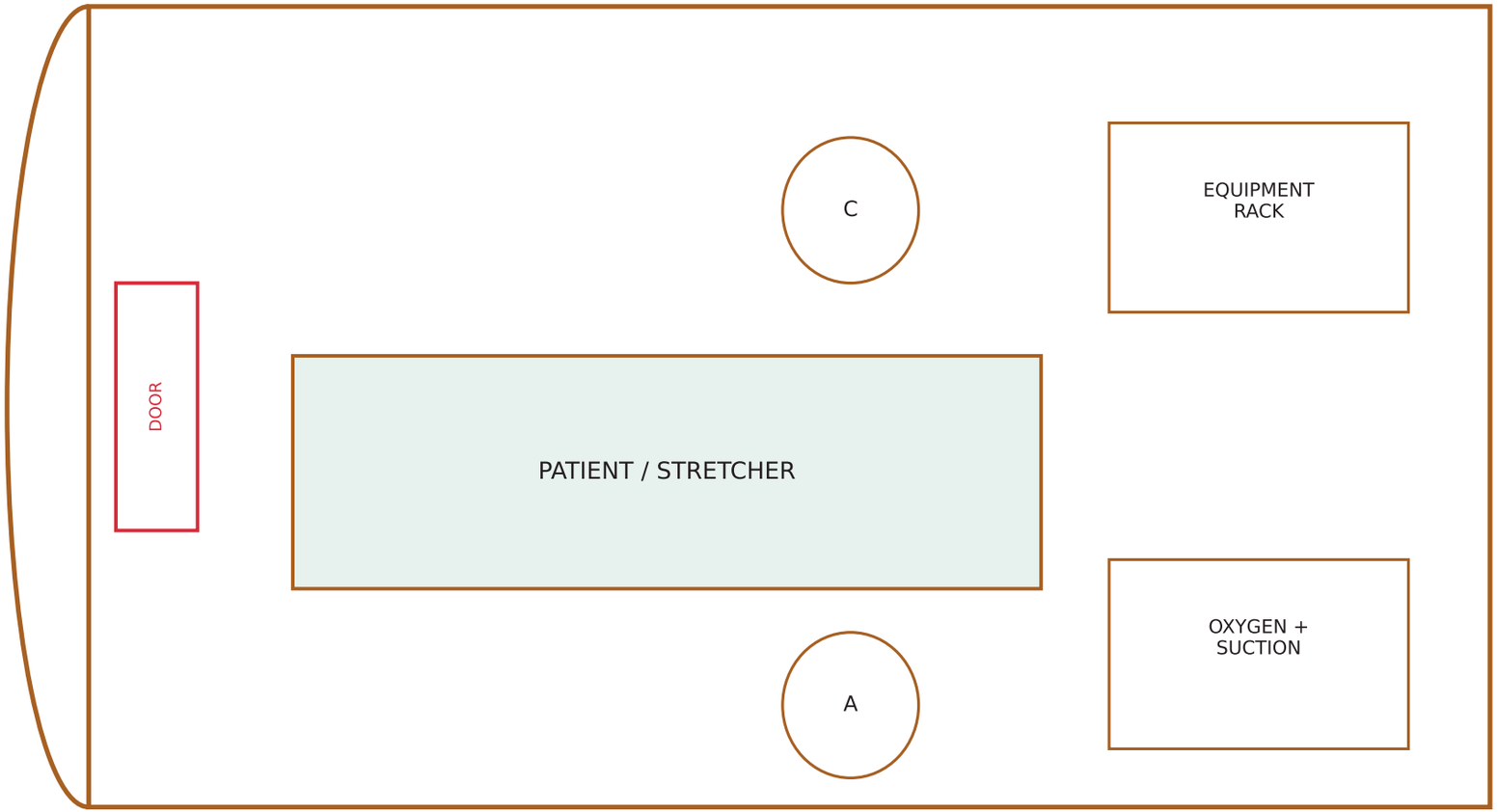
Exact fasteners, hinges, panel overlaps and wiring are unknown.

Assembly use

Subsystem STLs support independent edits in Tinkercad.

Proposed ambulance pod interior layout

Concept-only clinical layout inside a future pod envelope. Not validated for medical workflow, accessibility or regulation.



WORKING ZONES

- Patient transfer path
- Clinician reach zone
- Primary emergency exit
- Clinical equipment

Interior baseline

Stretcher centerline, clinician at side/head, assistant opposite.

Systems

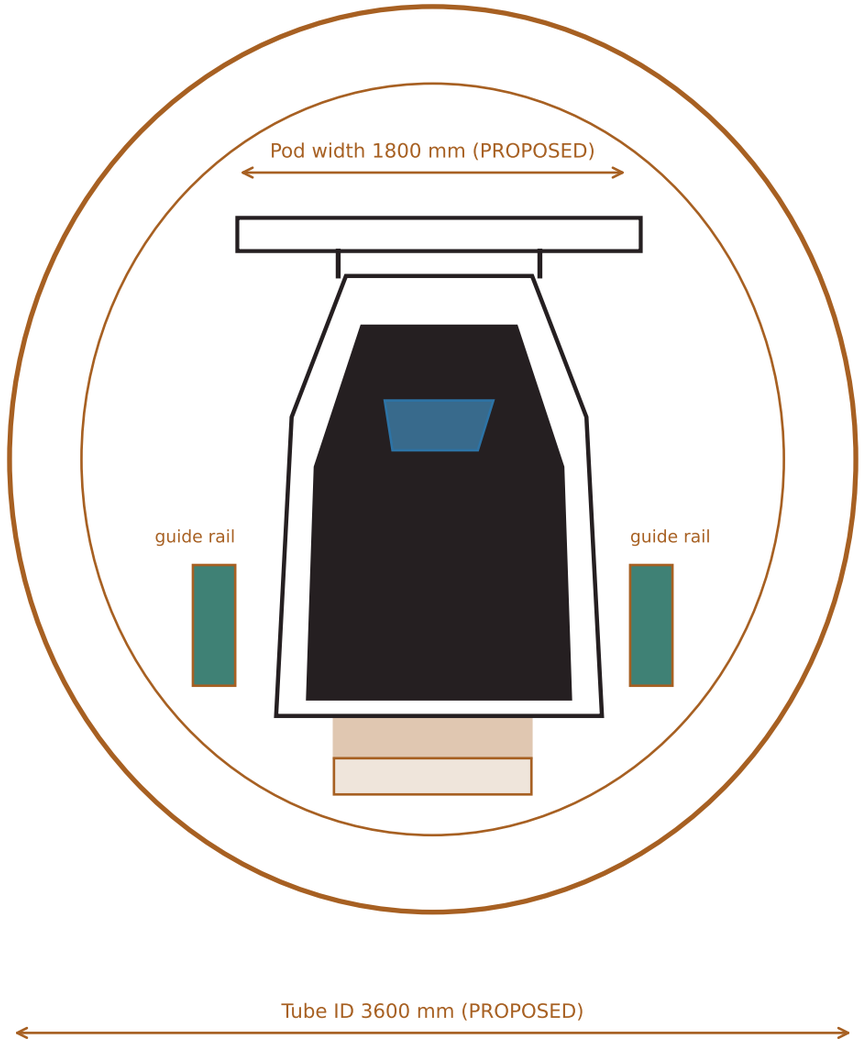
Monitoring, oxygen, suction, storage and crash-rated restraints.

Validation needed

Human factors, infection control, EMC, fire and evacuation.

Proposed tube and guideway interface

Concept section showing the ambulance pod inside a low-pressure transport tube. Interface dimensions remain open.



INTERFACE ZONES

- Pod primary structure
- Levitation air gap / guide reaction
- Linear propulsion reaction rail
- Service / rescue walkway
- Tube pressure boundary

Open engineering questions

Gap control, vacuum loads, rail tolerance, thermal growth, rescue access and debris protection.

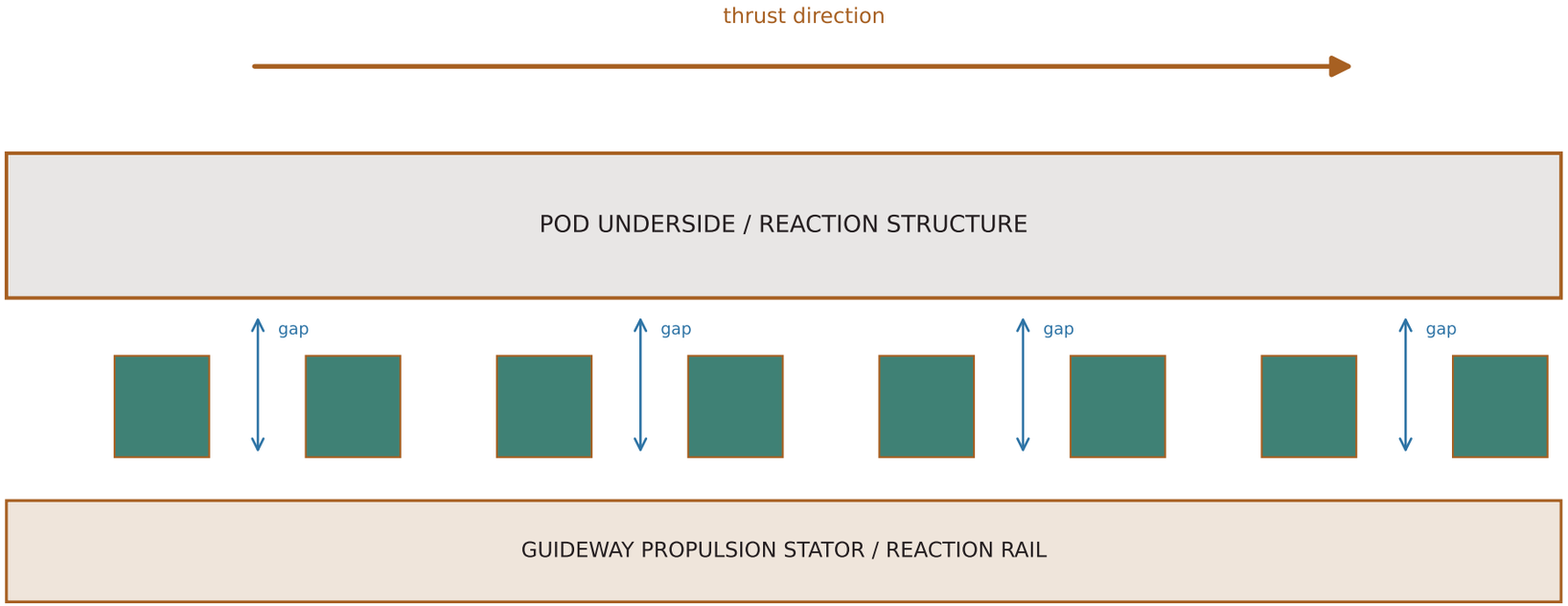
Levitation and propulsion zone diagram

Functional architecture only. Magnet topology, force, power, air gap and control laws are not specified.

LEVITATION

LATERAL GUIDANCE

PROPULSION



Control loop

Gap sensors -> vehicle controller -> levitation current.

Power

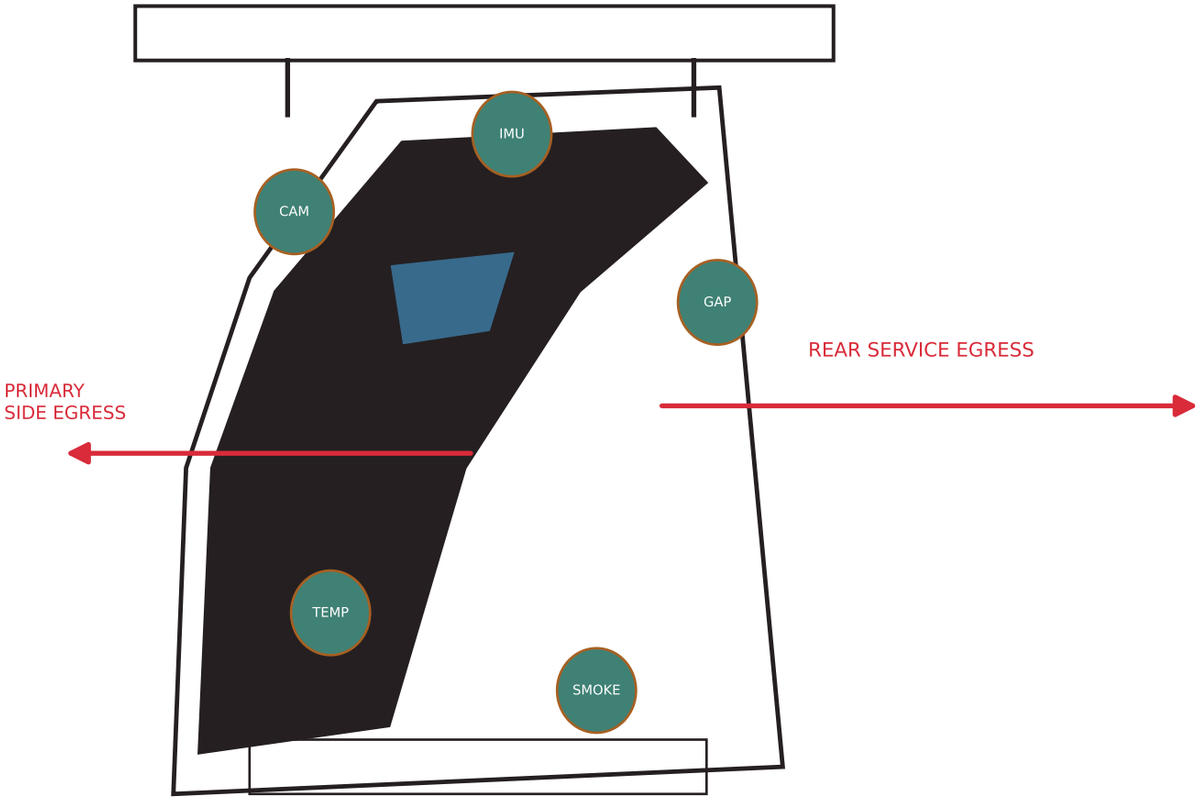
Segmented wayside power and isolated emergency coast/brake mode.

Fail-safe work

Define passive support, braking, thermal and fault containment.

Sensor placement and emergency egress

Proposed placement map with two independent egress directions. Final placement depends on structure and certification.



SENSOR SET

CAM	forward/rear visual
IMU	motion state
GAP	levitation clearance
TEMP	equipment thermal
SMOKE	fire detection
PRESS	cabin/tube pressure
BIO	patient monitor gateway

Emergency principle

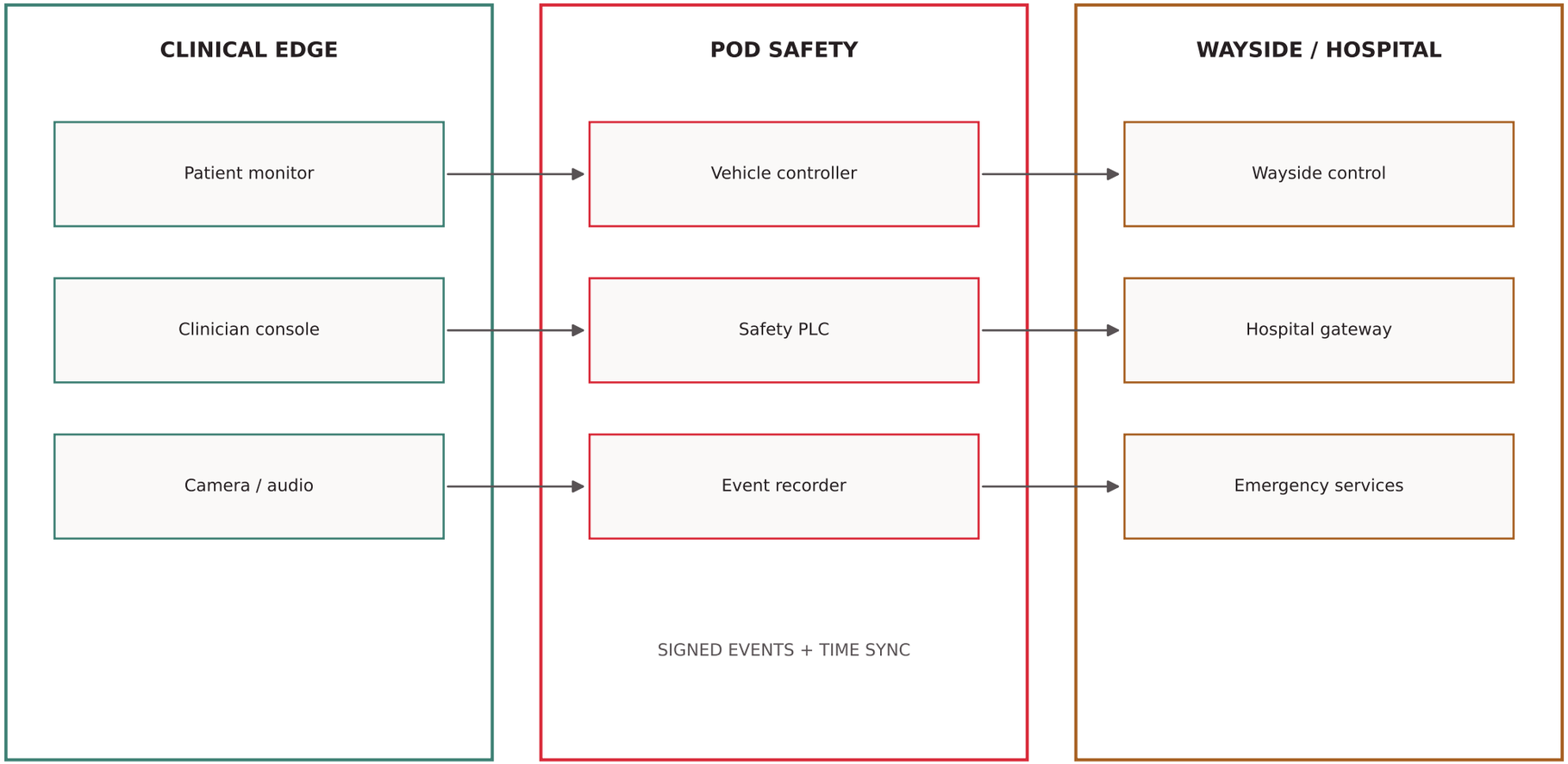
Manual release, photoluminescent path, independent power and clear rescue interface.

Verification

Egress time, door force, tunnel clearance, smoke spread and remote rescue.

Clinical and vehicle data-flow schematic

Logical separation between clinical data, safety control and transport operations. Cybersecurity boundaries are proposed.



Privacy boundary

Minimum necessary patient data; encrypted transport; role-based access.

Safety separation

Clinical services cannot command levitation, propulsion or braking.

Continuity

Offline local care remains available during network loss.

Integrated artifact index and revision map

Public artifact index for the mockup reconstruction, baseline concept model and this drawing set.

3D MODEL 01 | BASELINE CONCEPT

Tube, guideway and engineering overlays. Existing 1:30 package.

3D MODEL 02 | PHYSICAL MOCKUP REPLICA

Observed shell-focused 1:10 package. Merged STL, grouped OBJ/MTL, subsystem STLs, specs and previews.

2D ENGINEERING DRAWING SET

A01-A04 physical replica; B01-B05 proposed ambulance-hyperloop systems; B06 index.

STATUS KEY

OBSERVED

Directly visible geometry

ESTIMATED

Photo-derived dimension

PROPOSED

Engineering concept

Measurement revision

Replace photo-derived values after measuring W/L/H, window, bridge, bays, fans and doors.

Engineering revision

Carry requirements into CAD only after clinical, safety, tube and propulsion reviews.